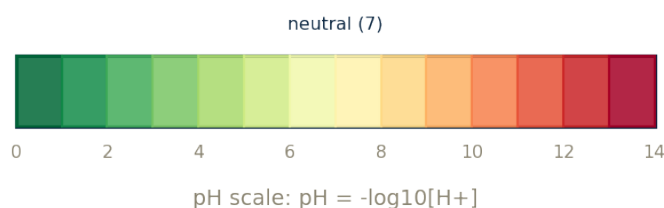


Ionic Equilibrium

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Arrhenius theory: acids produce H^+ ions and bases produce OH^- ions in water. Bronsted-Lowry theory: acids are proton donors, bases are proton acceptors.
- $pH = -\log_{10}[H^+]$; $pOH = -\log_{10}[OH^-]$; at 25 degrees C, $pH + pOH = 14$, and $[H^+][OH^-] = K_w = 10^{-14}$.
- Strong acids/bases dissociate completely in water; weak acids/bases dissociate partially, characterized by K_a or K_b .
- For a weak acid HA: $K_a = \frac{[H^+][A^-]}{[HA]}$. Degree of dissociation $\alpha = \sqrt{K_a/C}$ for small α (C = initial concentration).
- A buffer solution resists changes in pH upon addition of small amounts of acid or base; typically made of a weak acid and its conjugate base (or weak base and conjugate acid).
- Henderson-Hasselbalch equation: $pH = pK_a + \log\left(\frac{[salt]}{[acid]}\right)$, used to calculate buffer pH.
- Solubility product K_{sp} is the equilibrium constant for the dissolution of a sparingly soluble salt; if the ionic product exceeds K_{sp} , precipitation occurs.
- Common ion effect: adding a common ion to a solution suppresses the dissociation of a weak electrolyte (shifts equilibrium via Le Chatelier's principle).



The pH scale: $pH = -\log_{10}[H^+]$, ranging from acidic to neutral to basic.

2. Formula Sheet

pH	$\text{pH} = -\log_{10}[\text{H}^+]$
pH-pOH relation	$\text{pH} + \text{pOH} = 14$ (at 25 °C)
Ionic product of water	$K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$
Ka expression (weak acid HA)	$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$
Degree of dissociation	$\alpha = \sqrt{K_a / C}$
Henderson-Hasselbalch equation	$\text{pH} = \text{pK}_a + \log\left(\frac{[\text{salt}]}{[\text{acid}]}\right)$
Solubility product (A_xB_y)	$K_{sp} = [\text{A}]^x [\text{B}]^y$
pOH	$\text{pOH} = -\log_{10}[\text{OH}^-]$

3. Practice Problems (22 questions)

- Find the pH of a solution with $[\text{H}^+] = 10^{-3} \text{ M}$.
- Find the $[\text{H}^+]$ concentration of a solution with $\text{pH} = 5$.
- Find the pOH of a solution with $\text{pH} = 9$.
- A weak acid has $K_a = 1.8 \times 10^{-5}$ and initial concentration 0.1 M. Find its degree of dissociation.
- Find the pH of a buffer solution with $\text{pK}_a = 4.74$ and equal concentrations of acid and its conjugate base salt.
- Explain the Bronsted-Lowry definition of acids and bases, and give one example of a conjugate acid-base pair.
- Find the solubility product K_{sp} of AgCl if its molar solubility is $1.3 \times 10^{-5} \text{ mol/L}$ ($K_{sp} = s^2$).
- State the common ion effect and explain how it affects the solubility of a sparingly soluble salt.
- Find the pH of a 0.01 M strong acid (HCl) solution, assuming complete dissociation.
- Find the pH of a 0.001 M strong base (NaOH) solution, assuming complete dissociation.
- Explain what makes a good buffer solution, and give one common example.
- Find the $[\text{OH}^-]$ concentration of a solution with $\text{pOH} = 4$.
- State the difference between K_a and pK_a , and explain how a smaller pK_a relates to acid strength.
- For a salt AB_2 with solubility s , write the expression for its solubility product K_{sp} .
- Explain why the pH of pure water is 7 at 25 degrees C.
- A buffer contains 0.2 M acetic acid and 0.2 M sodium acetate (pK_a of acetic acid = 4.74). Find the buffer's pH.
- State Le Chatelier's principle as applied to the common ion effect on the dissociation of a weak acid.
- Find the pH of a solution formed by mixing equal volumes of 0.1 M HCl and 0.1 M NaOH (complete neutralization).

19. Explain the difference between a strong electrolyte and a weak electrolyte, with one example of each.
20. If the solubility product of a salt AB is 1×10^{-10} , find its molar solubility ($K_{sp} = s^2$).
21. State why adding NaCl (a common ion source) to a saturated AgCl solution decreases the solubility of AgCl.
22. Explain the significance of the ionic product of water (K_w) remaining constant at a given temperature, regardless of whether the solution is acidic, neutral, or basic.

Answer Key

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|--|---|
| 1. pH = 3 | 2. $[H^+] = 10^{-5} \text{ M}$ |
| 3. pOH = 5 | 4. $\alpha \approx 0.0134$ (1.34%) |
| 5. pH = pKa = 4.74 (since $\log(1)=0$ when $[salt]=[acid]$) | 6. Conceptual - acid donates a proton, base accepts one; e.g., HCl/Cl ⁻ is a conjugate acid-base pair |
| 7. $K_{sp} = (1.3 \times 10^{-5})^2 = 1.69 \times 10^{-10}$ | 8. Conceptual - common ion shifts dissociation equilibrium backward (Le Chatelier), decreasing solubility |
| 9. pH = 2 | 10. pH = 11 |
| 11. Conceptual - buffer needs a weak acid/base and its conjugate in comparable amounts; e.g., acetic acid + sodium acetate | 12. $[OH^-] = 10^{-4} \text{ M}$ |
| 13. Conceptual - pKa = $-\log(K_a)$; smaller pKa means larger K_a , meaning a stronger acid | 14. $K_{sp} = [A][B]^2 = s(2s)^2 = 4s^3$ |
| 15. Conceptual - in pure water, $[H^+] = [OH^-] = 10^{-7} \text{ M}$, giving pH = 7 | 16. pH = 4.74 (equal concentrations of acid and salt) |
| 17. Conceptual - adding a common ion (e.g., A ⁻ from a salt) shifts the weak acid's dissociation equilibrium backward, suppressing further dissociation | 18. pH = 7 (neutral, complete neutralization of strong acid and strong base) |
| 19. Conceptual - strong electrolyte dissociates completely (e.g., NaCl); weak electrolyte dissociates partially (e.g., CH ₃ COOH) | 20. $s = 1 \times 10^{-5} \text{ mol/L}$ |
| 21. Conceptual - added Cl ⁻ ions shift the $AgCl \rightleftharpoons Ag^+ + Cl^-$ equilibrium backward, reducing solubility (common ion effect) | 22. Conceptual - K_w is constant at fixed T; as $[H^+]$ increases (acidic), $[OH^-]$ decreases proportionally, and vice versa, keeping the product constant |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.